a

Methods and decision workflow

Features & supervised learning

Transaction features + Past-only history → Supervised model → Risk score

Training labels used for fitting

PaySim / IEEE-CIS: LightGBM, logistic regression
BankSim: LightGBM, XGBoost

Datasets

PaySim
IEEE-CIS
BankSim

Model scores

Scores & review decisions

score rank → Review selection → Review list → TP / FP / FN and metrics

Held-out labels

Threshold transfer or exact top-k
Decision rule specified per dataset

Past-only history is constructed without fraud labels; all miniature graphics are schematics.

b

Datasets

PaySim — 6,362,620 synthetic mobile-money transactions
Recipient history

IEEE-CIS — 590,540 labeled transactions
Card-related identifier history

BankSim — 594,643 simulated transactions
Merchant history

Three dataset-specific evidence settings
Source dataset sizes shown above

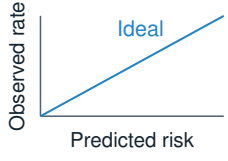
c

Metrics

Ranking & probability quality

AP (primary ranking)

Brier · log loss · ECE
Calibration analysis:
PaySim only



Review utility & analytical cost

TP · Recall
Precision
Actual alert rate



Analytical cost = FP + c × FN
Primary cost setting: c = 100 (PaySim / IEEE-CIS)

d

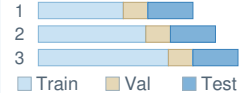
Evaluation design

Strictly past history



Earlier time groups only
Exclude the current group
No fraud-label history

Temporal validation
PaySim / IEEE-CIS



BankSim: one test window

Review decision rules
PaySim threshold rule:
Validation threshold → Test
Component audit: exact 3%

IEEE-CIS / BankSim:
Exact top-k (3% primary)

Uncertainty & sensitivity
Paired block bootstrap



Fitted models held fixed
Conditional intervals
Capacity / cost sensitivity